SmartTriage System

Patient Journey Flow and Safety Architecture

Version 1.0

February 18, 2026

Sections 1-3: Complete Clinical Workflow and Safety Enforcement

Contents

1 Complete Patient Journey Flow **2**
1.1 Phase 1 — Entry Point . . . 2
1.2 Phase 2 — Immediate Visual Assessment (0–60 seconds) . . . 2
1.3 Phase 3 — mSAT Triage Process . . . 3
1.4 Phase 4 — Waiting + Monitoring . . . 4
1.5 Phase 5 — Doctor Assessment . . . 4
1.6 Phase 6 — Investigations . . . 5
1.7 Phase 7 — Management . . . 5
1.8 Phase 8 — ICU (If Required) . . . 6
1.9 Phase 9 — Disposition . . . 6

2 Complete System Architecture Map **8**
2.1 SmartTriage Core Engines . . . 8
2.2 System Integration Diagram (Conceptual) . . . 8

3 How the System Enforces Safety **9**
3.1 Design Philosophy . . . 9
3.2 Enforce Protocol, Not Suggest . . . 9
3.3 Reduce Human Oversight Error . . . 9
3.4 Predict Deterioration Before Collapse . . . 9
3.5 Protect Pediatric & Infectious Cases . . . 10
3.6 Generate Structured Handovers . . . 10
3.7 Track Time-Sensitive Emergencies . . . 10
3.8 Safety Enforcement Summary . . . 11

1 Complete Patient Journey Flow

1.1 Phase 1 — Entry Point

Entry Types

- **Ambulance** — Emergency medical services transport
- **Walk-in** — Ambulatory patient presenting without prior notification
- **Referral** — Transfer from another facility with accompanying documentation

Step 1: Arrival Registration

System automatically records the following immutable data points:

- **Arrival Time** — Auto-generated, cannot be modified after creation
- **Entry Type** — Selected from predefined categories
- **Referring Facility** — Mandatory if referral type selected
- **Triage Nurse Assigned** — Responsible clinician for initial assessment

Critical Rule: Arrival Time is immutable and serves as the medico-legal start of care. Any corrections require administrative override with full audit logging.

1.2 Phase 2 — Immediate Visual Assessment (0–60 seconds)

Clinician performs rapid visual assessment at first contact. This occurs before formal triage for critically unstable patients.

Red Flag Criteria

If patient presents with ANY of the following:

- Unresponsive
- In respiratory distress
- Severe bleeding (hemorrhagic shock suspected)
- Active seizure
- Cyanotic

System Actions

- System **bypasses** standard triage module completely
- Patient automatically assigned **RED category**
- Patient directed to **Resuscitation Bed** (resus bay)
- **"Direct-to-Resus"** timestamp automatically recorded

No Override Allowed: This bypass cannot be disabled or overridden unless a senior clinician logs a mandatory justification explaining why the bypass was not appropriate.

1.3 Phase 3 — mSAT Triage Process

Step 1: Emergency Signs Check

System prompts mandatory checklist requiring clinician confirmation:

- Airway compromise
- Severe respiratory distress
- Shock signs
- Convulsions
- Coma (AVPU = P/U)
- Severe burns
- Hypoglycemia

Decision Point:

- If ANY emergency sign present → **Category RED**
- If NO emergency signs → Proceed to TEWS Calculation

Step 2: TEWS Calculation

Required Inputs:

- Mobility
- Respiratory Rate
- Pulse
- Systolic BP
- Temperature
- AVPU
- Trauma present (Y/N)
- SpO

System auto-calculates TEWS (Triage Early Warning Score) using Rwanda mSAT parameters.

Decision Rules:

- **TEWS 7** → Category RED
- **SpO $<$ 92%** → Category RED (regardless of other scores)
- **TEWS 5–6** → Category ORANGE
- **TEWS 0–4** → Proceed to Very Urgent Signs Check

Step 3: Very Urgent Signs

System displays separate medical and trauma checklists for very urgent criteria.

Decision Point:

- If ANY Very Urgent sign present → **Category ORANGE**
- If NO Very Urgent signs → Proceed to Urgent Signs Check

Step 4: Urgent Signs

System displays urgent criteria checklist.

Decision Point:

- If ANY Urgent sign present → **Category YELLOW**
- If NO Urgent signs → **Category GREEN**

System Auto-Assignment Rules

Category	Color	Max Treatment Time
RED	Red	Immediate
ORANGE	Orange	10 minutes
YELLOW	Yellow	30 minutes
GREEN	Green	60 minutes

Table 1: Triage Categories with Mandatory Treatment Timers

Note: *Triage Finish Time* is auto-recorded upon category assignment. This starts the treatment timer for the assigned category.

1.4 Phase 4 — Waiting + Monitoring

All patients, regardless of category, are connected to continuous monitoring devices where available.

System Functions During Waiting Period

- Continuously captures vital signs from connected monitors
- Recalculates rolling TEWS score automatically every 2 minutes
- Checks for deterioration trends using slope analysis
- Compares current status against category thresholds

Deterioration Response

If category breach detected OR worsening trend identified:

- **Auto-upgrade** patient to higher category
- **Alert triggered** to assigned clinician and charge nurse
- **Escalation logged** in audit trail with timestamp

1.5 Phase 5 — Doctor Assessment

Mandatory Documentation Fields

- Time doctor notified
- Time doctor attended
- History of present illness
- Allergies
- Risk factors
- Current medications
- Chief complaint
- Physical findings
- Working diagnosis

Automated Safety Checks

During documentation, system automatically verifies:

- Contraindications for any ordered medications
- Potential drug-drug interactions
- Sepsis screening criteria (prompts if not completed)
- Stroke/MI pathway eligibility (if symptoms suggest)

1.6 Phase 6 — Investigations

Order Types

- Laboratory tests (blood, urine, microbiology)
- Imaging studies (X-ray, CT, ultrasound, MRI)
- ECG / cardiac monitoring

Integration Features

- Orders transmitted electronically to respective departments
- Results auto-integrated into patient record upon completion
- Critical results **auto-flagged** with RED alert
- Pending results visible on patient dashboard

1.7 Phase 7 — Management

Management Components

- Medication administration log
- Procedures performed
- IV fluids (type, rate, total volume)
- Oxygen therapy (flow rate, delivery method)
- Ventilation settings (if ICU)

Medication Safety Requirements

Each medication entry requires:

- **Prescribed by** — Named clinician
- **Administered by** — Named nurse/clinician
- **Countersigned by** — Second clinician (mandatory for high-risk drugs)
- **Time** — Exact administration timestamp

1.8 Phase 8 — ICU (If Required)

ICU-Specific Documentation

- Ventilator parameters (mode, FiO, PEEP, TV, RR)
- Vasopressor tracking (drug, dose, rate, trend)
- Fluid balance (hourly input/output)
- Sedation score (RASS, Ramsay, etc.)
- GCS (Glasgow Coma Scale) hourly

Continuous Monitoring

- Deterioration prediction engine remains active
- ICU-specific alerts for hemodynamic instability
- Integration with vasopressor titration protocols

1.9 Phase 9 — Disposition

Three Possible Outcomes

1. Discharge to Home

- Auto-generated discharge summary
- Final diagnosis documented
- Discharge medications prescribed and reviewed
- Follow-up instructions provided

2. Admission to Hospital

- Structured SBAR handover generated automatically
- Receiving ward notified electronically
- Transfer time logged

- Pending investigations transferred with patient

3. External Transfer to Another Facility

- Comprehensive transfer summary generated
- Stability confirmation required before transfer
- Ambulance documentation completed
- Receiving facility acknowledgment obtained

Exit Process

- **Exit Time** auto-recorded upon disposition completion
- Patient record locked for further edits (except administrative corrections)
- Billing and reporting data finalized

2 Complete System Architecture Map

2.1 SmartTriage Core Engines

The SmartTriage system comprises interconnected modular engines, each responsible for specific clinical and operational functions.

- Entry & Registration Engine
- mSAT Triage Engine
- TEWS Calculator
- IoT Vital Integration
- Dynamic Re-triage AI
- Pediatric Logic Engine
- Sepsis Detection Engine
- Stroke/MI Fast Track
- Hypoglycemia Enforcement
- Infection Isolation Module
- Mental Health Protocol
- Clinical Documentation System
- Medication Safety Engine
- Lab Integration Engine
- Radiology Integration
- Deterioration Prediction Engine
- Continuous TEWS Graph Engine
- Clinical Pathway Automation
- Audit & Legal Engine
- ICU Ventilator Integration
- Vasopressor Monitor
- Fluid Balance Tracker
- Sedation & Neuro Scoring
- Structured Handover Generator
- Disposition & Reporting Engine

2.2 System Integration Diagram (Conceptual)

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                    SMARTTRIAGE CORE PLATFORM

  INPUT LAYER    PROCESSING          OUTPUT LAYER

Registration   Triage Engine   Category Assignment
Vitals (IoT)   TEWS Calculator Alerts & Notifications
Lab Results    AI Prediction   Clinical Documentation
Imaging        Pathway Logic   Handover Summaries
Medications    Safety Checks   National Reporting
```

3 How the System Enforces Safety

3.1 Design Philosophy

The SmartTriage system is built on a fundamental principle: **enforcement, not suggestion**. Safety is not optional—it is engineered into every workflow.

3.2 Enforce Protocol, Not Suggest

- **Mandatory field completion** before proceeding to next step
- **No manual triage category assignment**—system assigns based on objective data only
- **Cannot skip Emergency Signs step**—hard stop until checklist completed
- **Cannot discharge** without final diagnosis AND clinician signature
- **No override** without documented justification and senior approval

Hard Stop Example: If a clinician attempts to discharge a patient without completing the final diagnosis field, the system blocks the action and displays: "Discharge requires final diagnosis. Please complete assessment before proceeding."

3.3 Reduce Human Oversight Error

- **Auto-capture vitals** from connected monitors—eliminates transcription errors
- **Auto-calc TEWS**—removes mathematical errors and scoring inconsistencies
- **Auto-drug interaction checks**—alerts before medication reaches patient
- **Mandatory glucose** for altered mental state—prevents missed hypoglycemia
- **Weight-based dosing enforcement** in pediatrics—hard stop if dose exceeds safe range

3.4 Predict Deterioration Before Collapse

- **Trend analysis**—system evaluates patterns, not single values
- **Rate-of-change detection**—identifies rapid deterioration even before thresholds breached
- **TEWS rising trend alert**—notifies clinicians of worsening status
- **AI prediction**—identifies high-risk patients using machine learning models

Example: A patient with TEWS=4 (GREEN) shows RR increasing from 18→22→26 over 30 minutes. System detects rate-of-change >20% and upgrades to YELLOW with alert—before clinical decompensation.

3.5 Protect Pediatric & Infectious Cases

- **Age-adjusted vitals**—normal ranges change by age; system applies correct thresholds
- **Weight-based dosing**—mandatory weight entry; auto-calculation of safe doses
- **Isolation screening mandatory**—cannot complete triage without answering key questions
- **Pediatric-specific alert thresholds**—more sensitive for infants and children
- **TB/Measles screening**—automatic prompts based on presenting symptoms

3.6 Generate Structured Handovers

- **Auto SBAR format**—Situation, Background, Assessment, Recommendation
- **Pending labs highlighted**—visible to receiving team
- **Active drips listed**—medications, rates, start times
- **Recent deterioration flagged**—trend alerts visible in handover
- **Code status and advanced directives**—clearly displayed

3.7 Track Time-Sensitive Emergencies

Automatic Timers

The following timers auto-start based on clinical triggers:

Event	Target Time	Trigger
ORANGE category treatment	10 minutes	Category assignment
YELLOW category treatment	30 minutes	Category assignment
Stroke door-to-CT	20 minutes	FAST positive
MI door-to-ECG	10 minutes	Chest pain + risk factors
Sepsis 1-hour bundle	60 minutes	Sepsis alert

Table 2: Time-Sensitive Emergency Timers

Escalation Protocol

If any timer is breached:

- **Level 1 Alert** — Notification to assigned clinician (at 80% of target)
- **Level 2 Alert** — Notification to charge nurse/supervisor (at 100% of target)
- **Level 3 Alert** — Department-wide alert with escalation to clinical lead (at 120% of target)

Timer Example: ORANGE patient assigned at 14:00. System starts 10-minute timer. At 14:08 (80% time elapsed), clinician receives reminder. At 14:10, if doctor not attended, charge nurse notified. At 14:12, clinical lead alerted and patient auto-upgraded for safety review.

3.8 Safety Enforcement Summary

- **Prevent errors before they occur** — proactive design
- **Catch errors if they occur** — real-time monitoring
- **Learn from near-misses** — audit and analytics
- **Standardize best practices** — protocol enforcement
- **Protect vulnerable populations** — pediatric and infectious disease focus

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The SmartTriage safety architecture transforms passive documentation into active patient protection. Every module, every field, every alert is designed with one goal: prevent harm before it happens.